Analyzing Business Performance Using Demand Forecasting Techniques

Final Project Report

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SNT Summer Project'23

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Academic Year 2022–23

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1 Introduction

The retail industry generates large volumes of transactional data that can be used to understand customer behaviour, optimize operations, and improve profitability. Among the analytical tasks this data enables, *demand forecasting* — estimating future customer demand from historical sales — is one of the most valuable.

1.1 Motivation

Inventory mismanagement is a major source of retail inefficiency. When demand is *underestimated*, products run out, customers are lost, and revenue opportunities disappear. When demand is *overestimated*, excess stock accumulates, storage and wastage costs rise, and working capital is locked in unsold goods. Accurate forecasting therefore has a direct effect on both profitability and customer satisfaction. Retail demand is shaped by seasonality, holidays, regional demographics, and category-specific buying cycles, which makes forecasting challenging but high-value.

1.2 Problem Statement

The assignment uses four retail groups and seven city labels to study weekly sales. The records contain missing store–department weeks, recurring seasonal peaks, and an indicator for selected holiday weeks. The forecasting target is recorded sales value. The source lacks unit sales and stockout information, so underlying demand cannot be measured directly.

1.3 Objectives

1. Audit the source data and document gaps in weekly coverage.
2. Describe sales patterns by case group, city label, and department.
3. Propose hypotheses for the anonymous department codes.
4. Compare forecasting methods on aligned holdout and backtest dates.
5. Generate a 26-week aggregate sales projection.
6. Discuss what the results can support in a retail planning exercise.

1.4 Workflow

The pipeline proceeds as: data understanding → cleaning → exploratory analysis → department inference → time-series analysis → model development and evaluation → forecasting → recommendations. The remaining sections follow this order.

2 Dataset Description

The four assignment files contain 32,454 rows. Each row matches a record in the Walmart recruiting dataset after mapping the case store identifiers to the source identifiers. The four files correspond to Walmart stores 1–7, 12–18, 23–29, and 34–40. The names Bigg Bazar, D-Mart, Spar, and Seven Eleven and the Indian city names are labels used in the assignment. They do not identify the underlying businesses or store locations. The analysis covers 143 week-ending Fridays from 5 February 2010 through 26 October 2012. Its expected grid has 168 store–department series, of which 164 are fully observed.

Table 1: Dataset features.

Feature	Description
`Date`	Weekly observation date (week-ending Friday).
`Store`	Store identifier (1–7), mapped to a city.
`Dept`	Department identifier (1–6); names hidden.
`Weekly_Sales`	Recorded weekly sales value.
`IsHoliday`	Indicator for a selected source holiday week.
`Company`	Retail group label assigned for the case.

A `City` column is derived from the case store number (Table 2). The source uses anonymous department identifiers. Section 5 discusses possible interpretations of their sales patterns.

Table 2: Store-to-city mapping used in the analysis.

Store	1	2	3	4	5	6	7
City	Delhi	Mumbai	Kolkata	Lucknow	Bangalore	Indore	Jaipur

3 Data Cleaning and Preprocessing

Each file was checked for duplicate keys, Friday dates, and coverage by store and department. The expected case grid gives a consistent basis for comparing coverage.

3.1 Spar — Excess Departments

Store 7 included 8,597 valid source records for department codes outside the assignment's 1–6 scope. Those rows were excluded from the analysis.

3.2 Seven Eleven — Missing Weeks

The expected grid contains 167 absent store–department weeks in the Seven Eleven group (Table 3). Store 3 (Kolkata label), Department 6, has 19 observed Fridays and 124 missing Fridays across five separate gaps. In chronological order the gaps lasted 31, 39, 35, 11,

and 8 weeks. The same observations are absent from the Walmart reference. The cause of the gaps and the sales during them are unknown.

Table 3: Missing weekly records in Seven Eleven (total 167).

Store (City)	Dept	Missing weeks
3 (Kolkata)	5	30
3 (Kolkata)	6	124
4 (Lucknow)	6	8
5 (Bangalore)	6	5

3.3 Training Data and Missing Values

The primary forecast sums the same 164 complete series each week. In the 168-series sensitivity check, imputation is restricted to training data available at the forecast cutoff. Missing training sales are filled with the most recent observed sale for that series. For leading gaps, the median of observed peer stores in the same group and department on that Friday is used. Holdout actuals are never imputed. A full-panel actual is available only when all 168 observations are present for that Friday.

4 Exploratory Data Analysis

The retailer and city labels are used for descriptive comparisons within the case. The exploratory figures use the reconstructed panel. The holiday table below uses the 164 consistently observed series. Forecast accuracy is evaluated separately on date-aligned observations.

4.1 Company Performance

Aggregating sales by company gives a clear ordering: D-Mart leads (≈158M), followed by Spar (≈140M) and Bigg Bazar (≈135M), with Seven Eleven well behind (≈89M) (Figure 1).

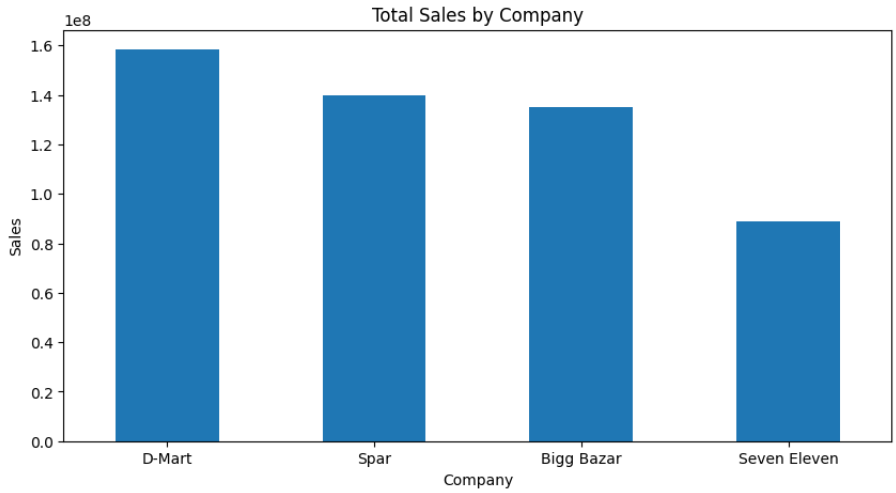


Figure 1: Total sales by company. D-Mart leads; Seven Eleven is the lowest by a wide margin.

Table 4: Approximate total sales by company over the full period.

Company	Total sales (approx.)	Share
D-Mart	158M	30%
Spar	140M	27%
Bigg Bazar	135M	26%
Seven Eleven	89M	17%

The group series share pronounced year-end peaks (Figure 2). This is consistent with a common seasonal pattern in recorded sales.

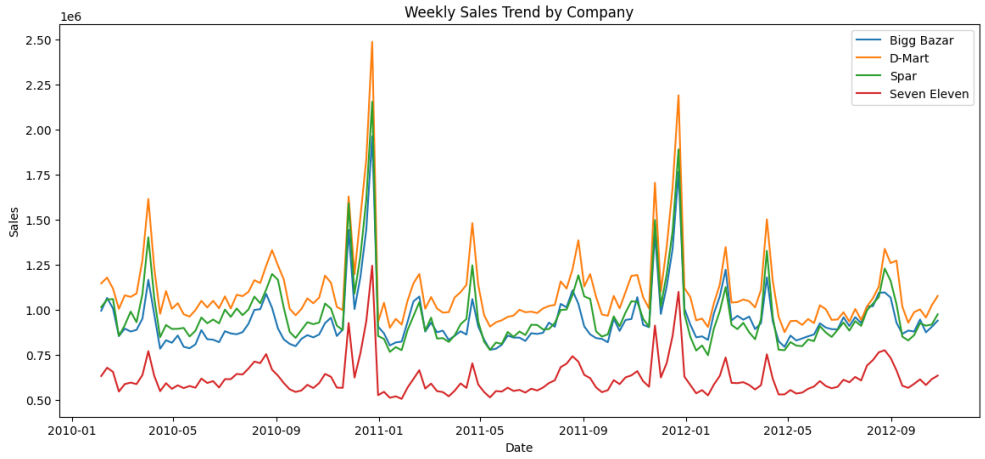


Figure 2: Weekly sales trend by company. The series are synchronized, with pronounced year-end spikes.

4.2 City and Regional Behaviour

Recorded sales are highest for the Mumbai label, followed by Indore and Delhi, and lowest for the Bangalore label (Figure 3).

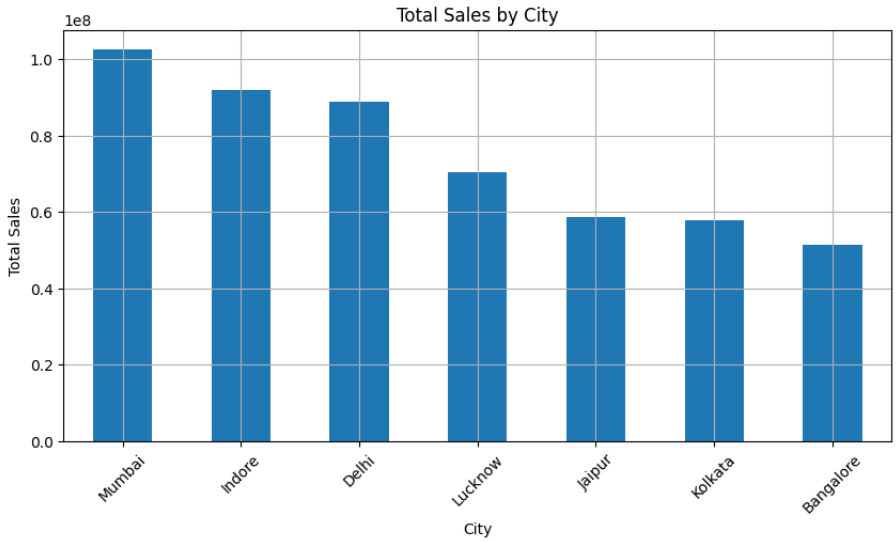


Figure 3: Total sales by city. Mumbai leads and Bangalore trails.

The highest-selling group varies across the city labels (Figure 4). Spar leads Bangalore and Delhi, Bigg Bazar leads Lucknow, D-Mart leads Mumbai, Kolkata, and Jaipur, and Seven Eleven leads Indore in the assignment data.



Figure 4: Sales by retail group and case city label.

Table 5 summarizes the highest recorded totals by case city label. Differences in coverage should be considered when comparing totals for these groups.

Table 5: Leading company by city (from Figure 4).

City	Leading company
Mumbai	D-Mart
Indore	Seven Eleven
Delhi	Spar
Lucknow	Bigg Bazar
Bangalore	Spar
Kolkata	D-Mart
Jaipur	D-Mart

4.3 Seasonality

The additive decomposition in Figure 5 separates trend, seasonal, and residual components in the exploratory series. It shows a pronounced year-end seasonal component. Forecast performance on that part of the calendar is assessed separately in Section 7.2.

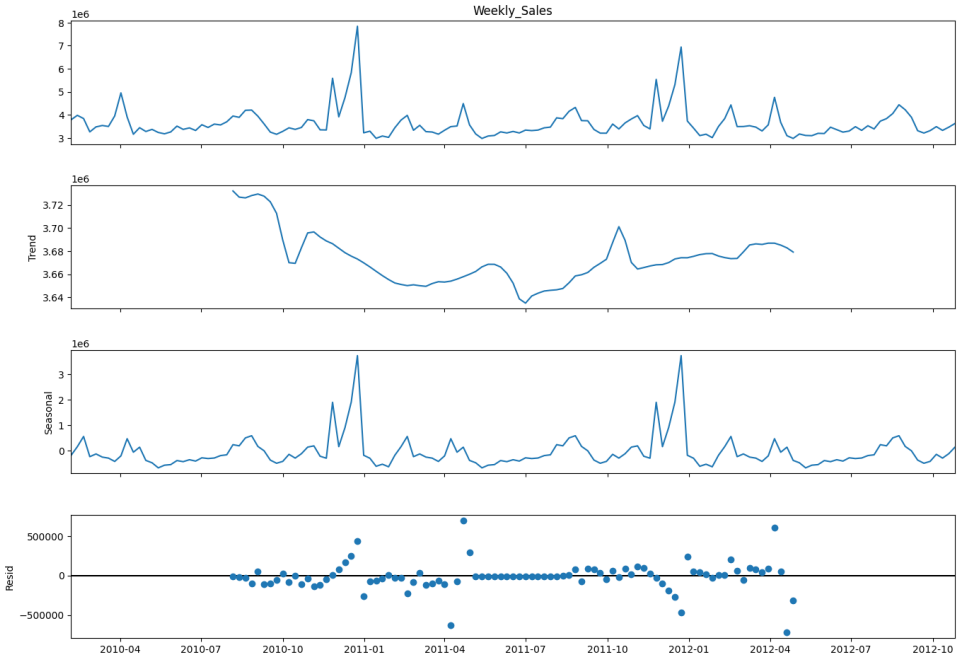


Figure 5: Additive decomposition of total weekly sales: weak trend, strong recurring year-end seasonality, and a small residual.

4.4 Holiday Associations

Average sales are higher on flagged Fridays in all four groups (Figure 6). The ten flags in this sample come from the Walmart competition calendar, which marks Super Bowl, Labor Day, Thanksgiving, and Christmas weeks. The comparison describes sales associations with these weeks. Season, promotions, and other factors may contribute to the observed differences.

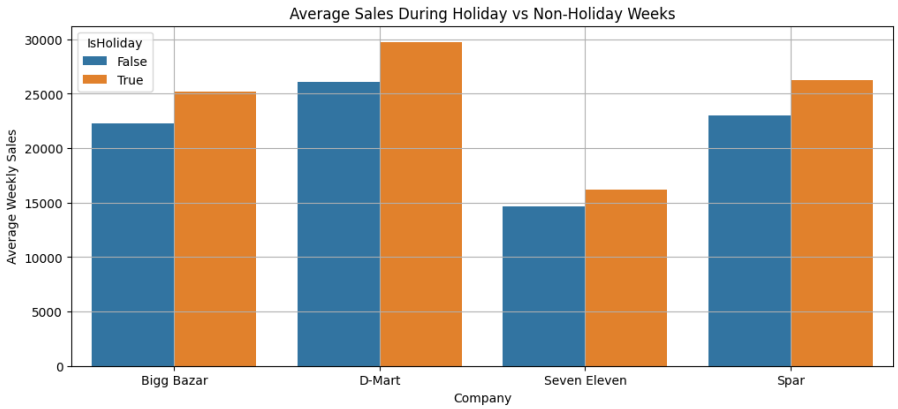


Figure 6: Average weekly sales during holiday vs non-holiday weeks, by case retail group in the exploratory panel.

Departments 5 and 6 had the largest differences on flagged Fridays in the complete cohort. Their mean sales were 57.33% and 75.26% higher, respectively. Department 2 averaged 2.88% lower (Table 6).

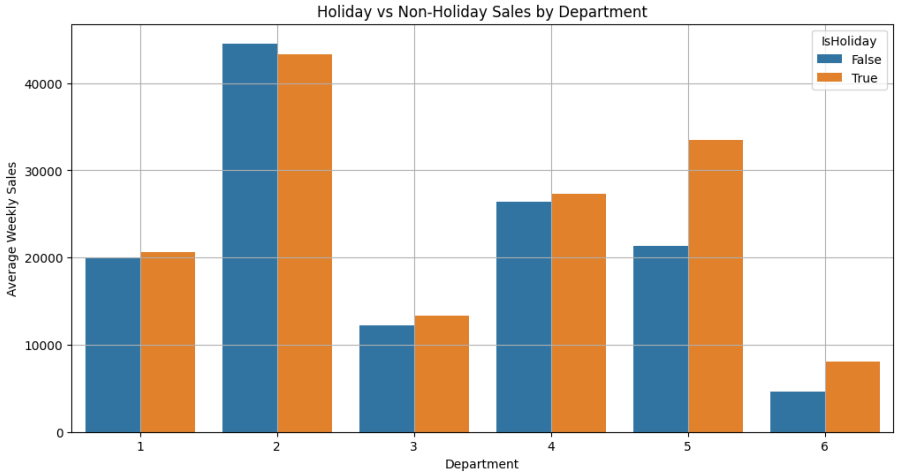


Figure 7: Mean weekly sales on flagged and other Fridays in the exploratory panel.

Table 6: Mean sales per store–department week in the complete cohort.

Department	Other Fridays	Flagged Fridays	Difference
1	20,002.42	20,677.39	+3.37%
2	44,552.65	43,270.17	−2.88%
3	12,248.17	13,361.91	+9.09%
4	26,415.20	27,275.38	+3.26%
5	22,086.70	34,749.36	+57.33%
6	5,152.94	9,031.23	+75.26%

4.5 Department Behaviour

Departments have different sales profiles (Figures 8 and 9). Department 2 has the highest mean sales, Department 3 peaks in August, and Departments 5 and 6 show pronounced year-end peaks.

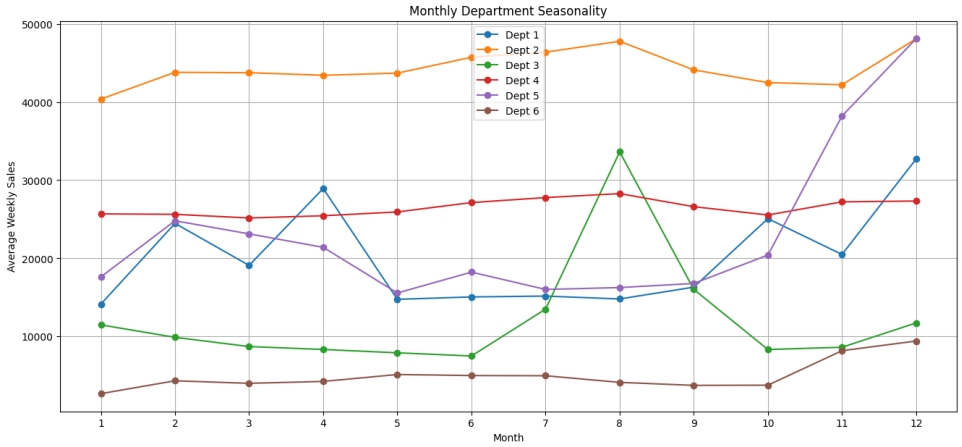


Figure 8: Monthly sales profiles by department code.

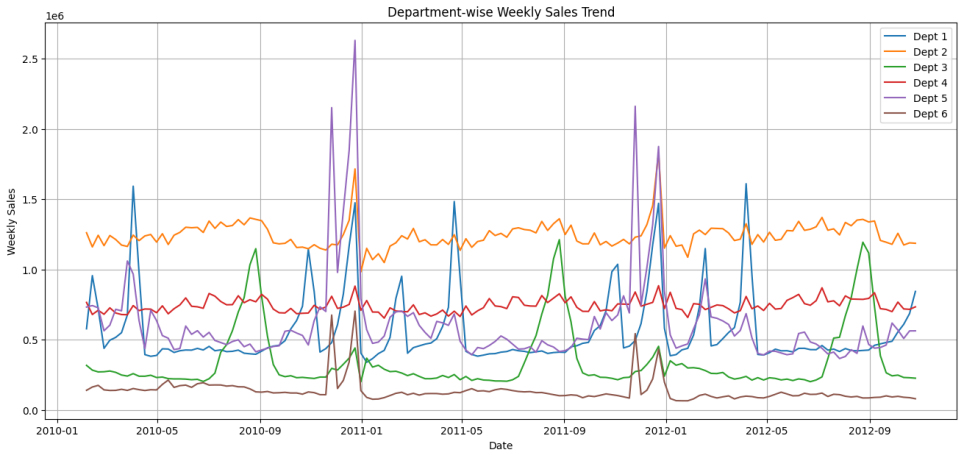


Figure 9: Weekly sales trends by department code.

5 Hidden Department Inference

The case asks whether sales levels, seasonal timing, and flagged-week associations suggest possible product categories. Table 7 records the hypotheses. Product labels are unavailable, and different categories can produce similar sales patterns.

Table 7: Department hypotheses and observed sales patterns.

Dept	Proposed category	Observed pattern
1	Clothing & Apparel	Repeated peaks and a modest flagged-week difference.
2	Groceries	Highest mean sales and a small negative flagged-week difference.
3	Books & Stationery	Highest mean weekly sales in August.
4	Beverages	Mid-range sales and a modest flagged-week difference.
5	Electronics	Year-end peak and a large flagged-week difference.
6	Furniture	Lowest mean sales and the largest proportional flagged-week difference.

6 Time-Series Analysis and Forecasting Methodology

Forecasts target total weekly sales value for the fixed cohort of 164 consistently observed series. Each prediction is matched to its actual by the exact week-ending Friday date. The holdout trains on 114 weeks through 6 April 2012 and forecasts the next 29 Fridays. The four earlier rolling cutoffs use 104, 106, 108, and 110 training weeks, with four forecasts per cutoff. Their 16 forecast–actual pairs cover ten distinct Fridays. The Augmented Dickey–Fuller test is computed on each training window as a diagnostic. Model orders are fixed across windows.

6.1 Models

Moving Average forecasts the next value as the mean of the last $k = 12$ training observations and repeats that value across the forecast horizon:

$$\hat{y}_{t+1} = \frac{1}{k} \sum_{i=0}^{k-1} y_{t-i}. \tag{1}$$

Exponential Smoothing (Holt–Winters, additive seasonality with period $m = 52$) maintains level, trend, and seasonal terms:

$$\ell_t = \alpha\,(y_t - s_{t-m}) + (1-\alpha)(\ell_{t-1} + b_{t-1}), \tag{2}$$
$$b_t = \beta\,(\ell_t - \ell_{t-1}) + (1-\beta)\,b_{t-1}, \tag{3}$$
$$s_t = \gamma\,(y_t - \ell_t) + (1-\gamma)\,s_{t-m}. \tag{4}$$

ARIMA$(2,0,2)$ combines autoregression and a moving-average term with backshift operator B:

$$\phi(B)\,(1-B)^d\,y_t = \theta(B)\,\varepsilon_t, \tag{5}$$

capturing short-range serial dependence around a constant mean.

SARIMA$(1,0,1)(0,0,1)_{52}$ adds a seasonal block of period $m = 52$ and was fitted on $\log(1 + \text{sales})$ to stabilise variance:

$$\phi(B)\,\Phi(B^m)\,(1-B)^d(1-B^m)^D\,y_t = \theta(B)\,\Theta(B^m)\,\varepsilon_t. \tag{6}$$

Prophet decomposes the series additively into trend and yearly seasonality:

$$y(t) = g(t) + s(t) + \varepsilon_t, \tag{7}$$

where $g(t)$ is a piecewise trend and $s(t)$ is a Fourier-series seasonal term. This specification does not include a separate holiday regressor.

Seasonal naive uses the observation 52 weeks before the target Friday. It benchmarks the fitted models against the prior year's weekly pattern.

6.2 Evaluation Metrics

Models were compared with Mean Absolute Error, Root Mean Squared Error, and Mean Absolute Percentage Error:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|, \quad \text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}, \quad \text{MAPE} = \frac{100\%}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right|. \tag{8}$$

MAE and RMSE are in sales-value units. RMSE weights larger errors more heavily. MAPE averages absolute percentage errors over nonzero actuals.

7 Results and Model Evaluation

Prophet had the lowest MAE, RMSE, and MAPE on the 29-week holdout. Seasonal naive had the lowest MAE and MAPE in the four-week rolling comparison, while ARIMA had the lowest rolling RMSE. These windows cover different forecast horizons and parts of the retail calendar.

Table 8: Forecast performance on the 164 consistently observed series.

Model	Holdout MAE	Holdout RMSE	Holdout MAPE	Rolling MAPE
Prophet	121,689.41	181,477.53	**3.58%**	7.04%
Seasonal naive	169,307.74	313,436.33	5.02%	**6.52%**
Holt–Winters	202,354.14	265,527.97	5.98%	8.01%
Moving average	330,996.41	381,015.37	9.69%	14.32%
ARIMA	383,009.66	426,570.21	11.28%	7.26%
SARIMA	1,096,160.22	1,147,789.02	32.58%	7.13%

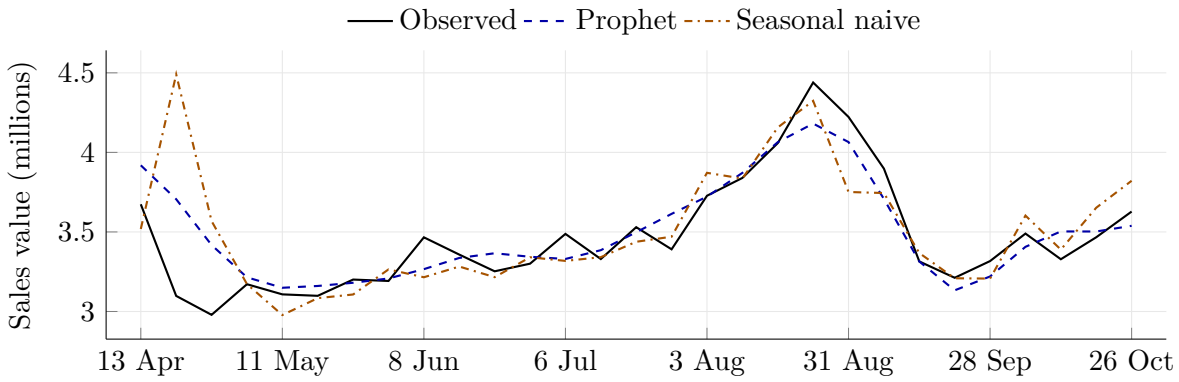


Figure 10: Observed sales and the two lowest-MAPE forecasts on the 2012 holdout.

7.1 Full-panel sensitivity

All six methods forecast the 168-series holdout. Only 8 June and 31 August 2012 had complete observed totals, and the rolling targets had none. Results on those two Fridays provide limited evidence about the full-panel target.

7.2 2011 year-end backtest

Prophet, seasonal naive, and the moving average trained on 91 weeks through 28 October 2011 and forecast the nine Fridays from 4 November through 30 December. These dates precede the 2012 holdout. The Thanksgiving observation was 5.54 million on 25 November. The observed maximum was 6.94 million on 23 December. Table 9 compares signed errors for Thanksgiving sales and the maximum forecast level.

Table 9: Year-end forecast accuracy and peak comparison.

Model	MAPE	Thanksgiving error	Peak date	Peak error
Seasonal naive	**7.03%**	+0.9%	23 Dec	+13.0%
Prophet	16.35%	−30.5%	16 Dec	−15.1%
Moving average	16.72%	−33.7%	All nine tied	−47.1%

Seasonal naive had the lowest MAE, RMSE, and MAPE and matched the observed peak Friday. Prophet placed the peak one week early and underpredicted Thanksgiving sales. This comparison covers one season.

8 Six-Month Forecast

Prophet was refitted through 26 October 2012 on the 164-series cohort and projected 26 Fridays from 2 November 2012 to 26 April 2013. Its highest point forecast is 5.72 million on 21 December 2012. Figure 12 shows the projection and the nominal 80% model intervals. The historical backtests evaluate point forecasts. Coverage of these prediction intervals has not been established.

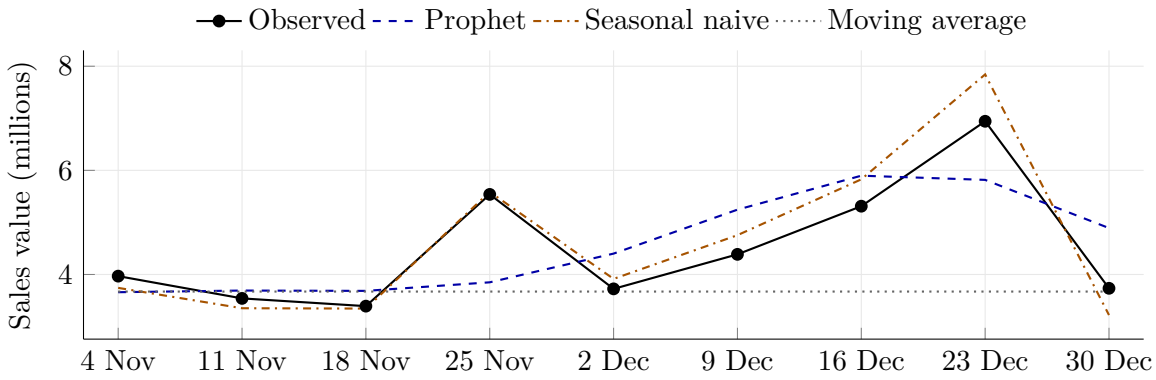


Figure 11: Observed and forecast sales during the 2011 year-end backtest.

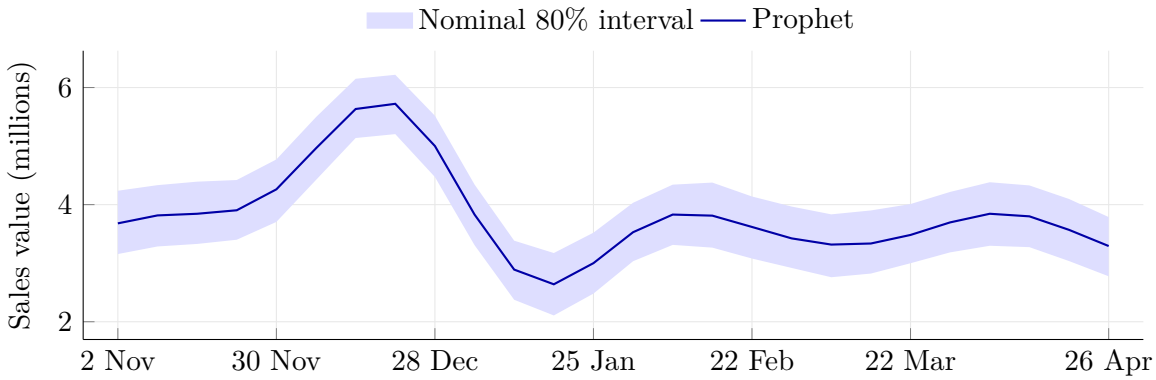


Figure 12: Prophet projection for the 164-series cohort, November 2012 to April 2013.

Table 10: Selected Prophet forecast dates, in millions of sales value.

Friday	Forecast	Lower bound	Upper bound
2 Nov 2012	3.68	3.16	4.24
23 Nov 2012	3.90	3.40	4.42
21 Dec 2012	5.72	5.20	6.22
18 Jan 2013	2.64	2.11	3.17
26 Apr 2013	3.29	2.78	3.79

9 Business Recommendations

1. **Benchmark forecasts by planning window.** Use seasonal naive as a reference for short-horizon and year-end forecasts. Compare peak timing and size alongside average errors when evaluating planning models.
2. **Review coverage before comparing groups.** Track absent store–department weeks and distinguish observed and imputed sales. Investigate the source of long gaps before relying on full-panel totals.
3. **Validate at the decision level.** Inventory and promotion decisions require actual product and store identifiers, unit demand, stockout records, and information on prices and promotions.

10 Conclusion

The analysis documents 167 missing store–department weeks, including 124 across five gaps in one series. Prophet led the 29-week holdout with 3.58% MAPE on the 164-series cohort. Seasonal naive led the four-week rolling comparison (6.52%) and the 2011 year-end test (7.03%). The year-end test shows why calendar coverage matters when choosing a model for a seasonal planning window.

The data support descriptive sales associations and department hypotheses. The year-end test covers one season, only two holdout Fridays have complete full-panel actuals, and the 26-week prediction intervals lack calibration. Further validation would require additional seasons and data at the store or product level.

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